Omkar Gudetti

Design Engineer | 3D CAD, Prototype Development & Product Lifecycle omkar.gudetti7@gmail.com
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PROFESSIONAL SUMMARY

Design Engineer with hands-on experience in 3D and 2D CAD design, prototype build and testing, engineering BOM and cost management, supplier coordination, and project delivery from concept to production. Proficient in Siemens NX, Creo, and SolidWorks, with directly transferable skills to Solid Edge given strong parametric CAD foundation. Experienced in producing detailed engineering drawings with correct dimensioning and tolerancing, leading small design projects from brief through implementation, building and testing prototypes, writing engineering reports, and working closely with manufacturing suppliers to resolve technical and production issues. Strong background in analytical problem solving and fault finding through structured root cause analysis. Working towards IMechE Chartered Engineer status.

CORE COMPETENCIES

- 3D CAD Design Siemens NX and Creo
- Engineering BOM and Cost Control
- GD&T and Tolerance Stack-Up Analysis
- Materials Selection and Heat Treatment
- Engineering Report Writing
- 2D Engineering Drawing and Detailing
- Supplier Coordination and Management
- FEA Structural and Thermal Analysis
- Design for Manufacture and Assembly
- Cross-functional Team Collaboration
- Prototype Build and Debugging
- Concept to Production Project Delivery
- Root Cause Analysis and
- Validation Testing and Test Reports
- Machinery and Equipment Design

PROFESSIONAL EXPERIENCE

GE Vernona | Mechanical Engineering Intern
Lisburn, UK Aug 2025 – Present

- Produced 3D CAD models and detailed 2D engineering drawings for electromechanical assemblies and components in Siemens NX, applying correct dimensioning and tolerancing principles and maintaining engineering BOM and configuration control within Teamcenter PDM
- Analysed large validation datasets from environmental and thermal testing to identify performance trends, anomalies, and root causes.
- Designed and built test rigs and instrumentation setups for electromechanical product validation, personally building and debugging test procedures, identifying faults through systematic analysis, and iterating on the setup to maximise data quality
- Conducted Ansys CFD thermal simulation of the DGA900 enclosure, writing formal technical reports documenting findings and design recommendations for internal and customer stakeholders
- Led fault finding and root cause analysis on a product failure mode identified during cold-weather validation testing to minus 50 degrees Celsius, implementing corrective actions that resolved a product shipment hold and improved overall product performance
- Coordinated with manufacturing and supply chain partners to resolve technical and production issues, applying DFM principles to ensure designs were producible within cost and schedule targets

Osler Diagnostics | Process Engineering Technician
Oxford, UK May 2025 – Aug 2025

- Designed and built validation test setups for precision sealing interface assemblies, personally running pressure, leak, and functional tests, recording results, and writing formal test reports against defined acceptance criteria
- Identified root cause of 94 percent component reject rate through tolerance stack-up analysis and fault finding, implemented corrective design and process changes, and verified the fix through follow-on testing, reducing reject rate to near zero and commissioning time by 20 percent

Oxford Brookes Racing | Powertrain Design Engineer
Oxford, UK Sep 2023 – Sep 2024

- Designed and detailed high-performance rotating mechanical components in Siemens NX and Creo, producing 2D engineering drawings with correct GD&T, managing BOM documentation, and coordinating with manufacturing suppliers to resolve design and production issues
- Built and tested prototype components on test rigs, performing structural and thermal FEA in Ansys to validate designs against performance requirements, and correlating simulation results against physical strain gauge test data to close the loop between analysis and hardware
- Led small design improvement projects from brief through to implementation, writing engineering reports documenting design decisions, analysis results, and test findings

Stallion Motorsport Electric | Design and Testing Head

Pune, India *Aug 2020 – Jun 2023*

- Personally, designed all primary mechanical components from concept to production in Siemens NX, including planetary gearbox housing, wheel upright, hub assembly, and HV battery enclosure, producing full 2D drawing packages with GD&T and managing engineering BOM and cost throughout the design programme
- Built and debugged prototype assemblies through multiple design iterations, specifying materials and surface treatments for rotating and structural components, applying DFM principles, and coordinating with external manufacturing suppliers to resolve technical and producibility issues
- Designed and fabricated the FSAE chassis, validating design through rigorous testing to ensure structural integrity and FSAE compliance.
- Led fault finding investigation on a gearbox overheating failure identified during the test programme, used systematic thermal data analysis to identify root cause, redesigned the affected component, revalidated on track, and confirmed zero recurrence before competition launch
- Led small to medium engineering projects from concept through to production hardware across chassis, aerodynamics, electronics, suspension, and brakes disciplines, managing suppliers, coordinating deliverables, and writing engineering documentation throughout, achieving All India Rank 1 at Formula Bharat 2023

BMW Mini Plant | Quality Inspector (Part-time)

Oxford, UK *Nov 2024 – Mar 2025*

- Reviewed components against engineering drawings and 2D specifications to ISO 9001 and Built-In Quality standards, raised non-conformances, and contributed to corrective action and continuous improvement processes on a high-volume automotive production line

EDUCATION

MSc Motorsport Engineering | 2:1 Honours | Oxford Brookes University, UK *2023 – 2024*

BEng Mechanical Engineering | First Class Honours | Savitribai Phule Pune University, India *2020 – 2023*

Diploma in Mechanical Engineering | First Class Honours | AISSMS, Maharashtra, India *2017 – 2020*

COURSES AND CERTIFICATIONS

Six Sigma Foundations | Simulation Using Ansys | Engineering Design | MATLAB | Python | HyperMesh | LS-DYNA | Diploma in Mechatronics | Electronics Foundations | Solid Edge (currently developing)

KEY PROJECTS

A Study on the Discrepancy in Applied Impact Forces on Passengers in Car Accidents, 2024

- Conducted crash simulations of a Toyota 2011 model using LS-DYNA and Oasys Primer, analysing passenger impact forces and proposed seat belt design improvements to enhance safety

Passenger Vehicle Aerodynamic Behaviour, 2024

- Performed FEA and CFD computations to analyse the aerodynamic behaviour of the Tesla Model 3, focusing on enhancements, crosswind performance, and the impact of wheel design

Design and Analysis of Drivetrain of FSAE Race Car, 2022 to 2023

- Developed and tested a dual-motor transmission with a custom-built Limited Slip Differential and gearbox to enhance traction and performance while ensuring system integration

Design of Planetary Gearbox, 2018 to 2019

- Developed planetary gearbox in partnership with Ratna Gear Industries, emphasising cost reduction and efficiency improvements while utilising CNC manufacturing for precision production

AWARDS

- Formula Bharat SAE 2022 to 2023: All India Rank 1 — led complete concept to production mechanical design programme and 532 km validation test campaign
- Formula Bharat SAE 2021 to 2022: All India Rank 8
- GE Vernova Workhuman Recognition 2026 — awarded by Senior Engineer for test instrumentation design, fault finding contribution, and engineering documentation supporting product overheating investigation
- Techno Vision 2020: Runner-up, presented paper on Hybrid and Electric Cars